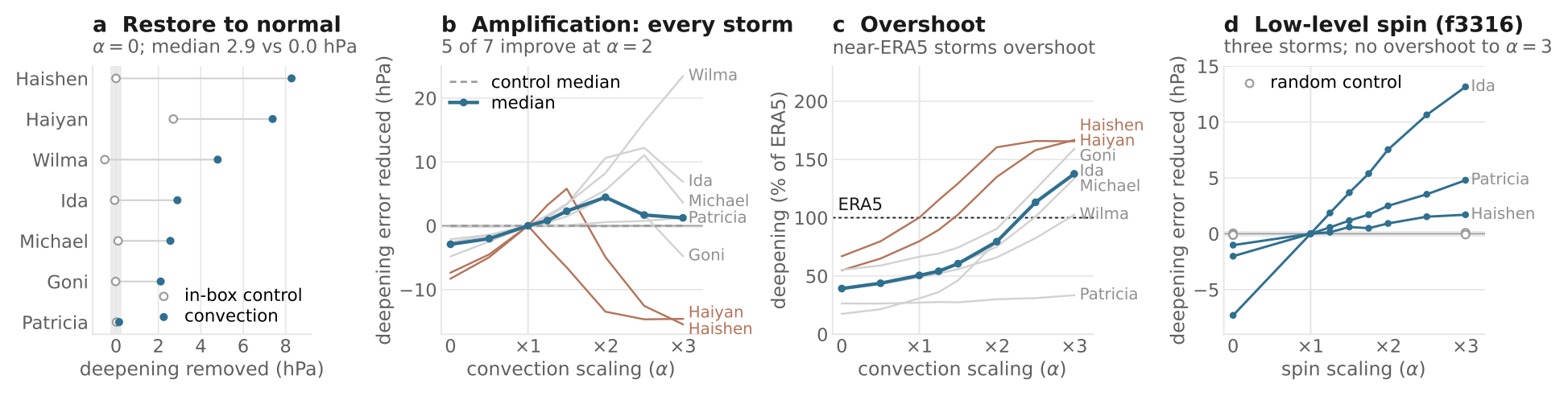
a Restore to normal
α = 0; median 2.9 vs 0.0 hPa
Haishen
Haiyan
Wilma
Ida
Michael
Goni
Patricia
in-box control
convection
0
2
4
6
8
deepening removed (hPa)
b Amplification: every storm
5 of 7 improve at α = 2
control median
median
deepening error reduced (hPa)
20
10
0
−10
Wilma
Ida
Michael
Patricia
Goni
Haiyan
Haishen
0
×1
×2
×3
convection scaling (α)
c Overshoot
near-ERA5 storms overshoot
deepening (% of ERA5)
200
150
100
50
0
ERA5
Haishen
Haiyan
Goni
Ida
Michael
Wilma
Patricia
0
×1
×2
×3
convection scaling (α)
d Low-level spin (f3316)
three storms; no overshoot to α = 3
random control
deepening error reduced (hPa)
15
10
5
0
−5
Ida
Patricia
Haishen
0
×1
×2
×3
spin scaling (α)